21-242 Recitation 12

Fields and Vector Spaces — Q5. =- need not imply =

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Show that $\vec{v} = -\vec{v}$ does not necessarily imply $\vec{v} = \vec{0}$.

Course-level solution

The equation $\vec{v} = -\vec{v}$ is equivalent to $2\vec{v} = \vec{0}$. If $1 + 1 \neq 0$ in $\mathbb{F}$ (characteristic not 2), we may multiply by the scalar 2^{-1} and conclude $\vec{v} = \vec{0}$. So a counterexample must live in characteristic 2.

Take $\mathbb{F} = \mathbb{F}_2 = \{0, 1\}$ with $1 + 1 = 0$, and take $V = \mathbb{F}_2$ as a vector space over itself. Let $\vec{v} = 1$. Then

$$-\vec{v} = (-1)\vec{v} = 1 \cdot 1 = 1 = \vec{v},$$

but $\vec{v} \neq \vec{0}$.

(The same phenomenon occurs in any vector space over a field of characteristic 2: every vector satisfies $\vec{v} = -\vec{v}$.)

Later-course solution (using ideas from the rest of the handout)

Once you have bases: if $\operatorname{char} \mathbb{F} = 2$ and $\dim V \geq 1$, any nonzero basis vector is a counterexample. If $\operatorname{char} \mathbb{F} \neq 2$, then $\vec{v} = -\vec{v}$ really does imply $\vec{v} = \vec{0}$, so the hypothesis "not necessarily" is a characteristic-2 statement, not a dimension statement.